



Aliens probably?

The analysed Kepler data points towards the possibility of existence of aliens nearer to us.
<http://dgit.in/NasaAlien>



Now lab grown Burgers!!!

After human organs now it's time for artificial lab grown human food. First is burger.
<http://dgit.in/LabGrown>

Tomorrow's tech

science that really has the potential to revolutionise the whole world. At the moment, it's really just a laboratory but if it's scalable into something that everyone can do, it could not only change the products that we make but how much electricity we need, pollution, infrastructure, almost every area of people's lives. Everybody could stop thinking about electricity generation as a power station and start thinking about it on an individual level, we're all generating electricity and that is genuinely very exciting.

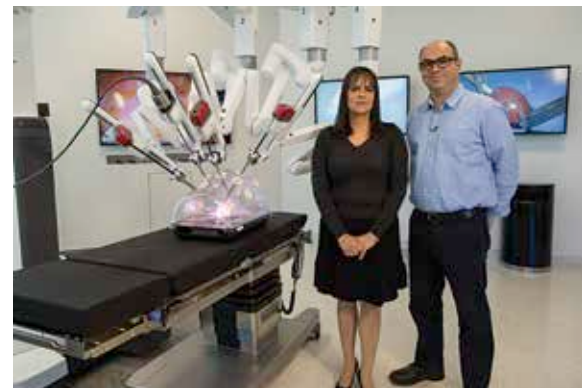
Robot surgery

The world of medical technology is also experiencing vast changes and this series we visit a facility in California where surgeons are being trained to use robots in surgery. The robot itself does not actually conduct the surgery, it is robot assisted; it's a bit like the game at the funfair where you have to control the pincers and grab a teddy bear. At the funfair they purposefully make it difficult to grab the teddy bear, here of course they make it much easier; to move the cutting arm, or the pincers a tiny amount you have to move your arm a lot, which means you have very precise control. This is something that is perhaps harder when doing surgery in person as moving the tiniest amount can be very difficult, but using robots you can control a millimetre by moving your arm a foot and the machine just scales that down.

I was allowed to have a go on the machinery and did the test that surgeons have to do. I had to control two mechanical arms through a series of gears, pick up tiny rubber bands and then stretch them over

tiny bubbles; I mastered the movement within an hour. What was interesting was that the operating table was only 2 or 3 feet away from me and the machine was controlling the surgeon's tools remotely, which to me suggested that there was no reason why a surgeon in California couldn't be operating on someone in London.

This technology really raises the possibilities for new kinds of treatment as expert surgeons could operate on patients that are on the other side of the world. This isn't something that's in a laboratory, this is really happening now and these expensive machines are being installed in hospitals all over the world. But this new frontier in surgery isn't about replacing surgeons, as there is always a surgeon the other side of the machine, making all the decisions, it's about giving surgeons a new set of technological tools and pushing the frontier of surgery even further forward.



Myriam J. Curet and Adam Shaw with the robotic arms that will enable the next generation of intuitive surgery

Frontier farms

We are now pushing at the frontiers of farming, not just growing crops on the edges of our productive land but right in the heart of our cities. This is an attempt to provide us with nutritious, locally produced food but also to revitalise an industry that is suffering from a declining and ageing workforce. When we think about frontier farms it conjures up images of farming in the wildest of places and in some cases, the opposite is true; it's actually about urban agriculture.

There are two places that are a good example of this; one is in Clapham, South London, where they have converted an old bomb shelter into an underground farm with polytunnels and a hydroponic system to grow fruit and vegetables. This particular venture is being partly funded by the chef, Michel Roux, who uses their herbs and salads

lettuce hydroponically in an old floppy disk factory. They use an old floppy disk factory because it is dust free and because there are no pests, animals or disease involved there is no need for any pesticide. They also control the environment and you have to wear a mask and pass through UV lights and vacuum chambers to make sure no contaminants get into the room. These factories of food are unlike any other farms and operate like an ordinary factory process.

So there's this irony or paradox, that the frontier is much closer than you think it is. You think of the frontier as somewhere in the wild but actually, the frontier in farming is bringing it home. 



Anna Holligan and Pavlos Kalaitzoglou, Wageningen: Building Frontier Farms

About the author: Award-winning journalist Adam Shaw is the host of Horizons, a BBC TV series that focuses on discovering the latest technologies and innovations that could revolutionize the way we live. During the first few seasons of Horizons, Adam has filmed in 18 different countries, visited 120 different companies and organisations and interviewed 150 industry and thought leaders travelling the globe searching for the brilliant ideas that have the potential to change millions of lives.